of $\mu = 0$Å and $\sigma = 0.2$Å. Similarly, isotropic and anisotropic cell deformations were sampled with a $\mu = 1\%$ and $\sigma = 4\%$. All structures along the relaxation trajectory were included in the dataset.

We note that these strategies were chosen to increase diversity, but there are many possible sampling strategies that could be used to maximize information content [64,65], and we expect these strategies to be useful for future sampling efforts. Active learning [10,24] sampling strategies have the potential to further enhance these approaches but it remains unclear how they compare to random baselines when considering large scale dataset sizes.

4.2 OMat24 DFT calculation settings and details

DFT calculations generally followed Material Project default settings [19] with some important exceptions. The calculations in this work have been performed using the ab-initio total-energy and molecular-dynamics package VASP (Vienna ab-initio simulation package) developed at the Institut für Materialphysik of the Universiät Wien [66,67] with periodic boundary conditions and the projector augmented wave (PAW) pseudopotentials [68]. Exchange and correlation effects were calculated using the generalized gradient approximation and the Perdew-Burke-Ernzerhof (PBE) with Hubbard U corrections for oxide and fluoride materials containing Co, Cr, Fe, Mn, Mo, Ni, V, or W, following Materials Project defaults [19].

VASP input sets were generated using the MPRelaxSet class defined in the pymatgen [69] library with the following modifications to account for recent updates and changes in the underlying algorithms and pseudopotentials:

1. Version 54 of pseudopoentials provided by VASP were used, rather than the legacy PBE MPRelaxSet defaults [19]. The Yb_3 and W_sv pseudopotentials were used for Yb and W to account for changes between version 52 and 54 of VASP PBE pseudopotentials [70,71].
2. All calculations were done with the ALGO flag set to "Normal".

A list of all differing pseudopotentials POTCAR symbols and generation dates is given in the Supplementatary Section A.

Relaxations were conducted with the MPRelax set defaults of VASP input flags. All AIMD calculations were carried out for 50 steps at a time-step of 2 femtoseconds. We note that 2 fs is a large time-step for typical AIMD simulations, especially for hydrogen-containing materials, but was chosen as the goal was to sample diverse configurations rather than perfectly integrate the trajectories. Finally, for static calculations resulting from inputs generated using rattling and Boltzmann sampling, only the NSW and IBRION flags where updated as appropriate for single point calculations.

4.3 OMat24 MLIP training strategies

We used the OMat24 dataset along with the MPtrj [16] and Alexandria [20] datasets to train GNN MLIPs. Since similar structures exist in the Alexandria and the WBM dataset used for testing, we subsampled the Alexandria dataset for training to ensure there was no leakage between the training and testing datasets. The new subset of Alexandria (sAlexandria) was created by removing all trajectories in which any structure matched a structure in the WBM initial and relaxed structures using structure prototype labels [35]. Next, we reduced the size of the Alexandria dataset by removing all structures with energies > 0 eV, forces norm > 50 eV/Å, and stress > 80 GPa. Finally, we only sampled structures in the remaining trajectories that had a energy difference greater than 10 meV/atom. The resulting datasets used for training and validation had 10 million and 500 thousand structures respectively.

We train and evaluate EquiformerV2 models [26] as it is currently one of the top performing direct force models on the OC20 [1], OC22 [72] and ODAC23 [4] benchmarks. We also train and evaluate eSEN models, since it is an energy conserving architecture which has shown SOTA performance for phonon and thermal conductivity calculations [28]. The models are trained to predict energy, forces and stress given an input structure. We explored three strategies:

1. EquiformerV2/eSEN models trained solely on the OMat24 dataset, with and without denoising augmentation objectives.

2. EquiformerV2/eSEN models trained solely on the MPtrj dataset, with and without denoising augmentation objectives, useful for direct comparison on the Matbench Discovery leaderboard (denoted “compliant“ models).

3. EquiformerV2/eSEN models from (1) or OC20 checkpoints further fine-tuned on either the MPtrj or sAlexandria datasets, leading to the highest performing models for the Matbench Discovery leaderboard (denoted “non-compliant“).

Additional model specifications, training information, and parameters are given in Supplementary Section G.

4.4 MLIP evaluations of energy, force and phonon softening

To evaluate systematic softening in model predictions, we predicted energy and forces for the $\tilde{1}000$ structures in the WBM high energy states dataset[23], and phonon frequencies for approximately 10,000 structures in the PBE MDR phonon dataset[49]. The zeroth order (energy) softening was computed as the difference in the predicted energy per atom with the DFT-calculated energy per atom. First order (force) softening is computed as the slope of the least squares fit of the predicted forces and the target forces[23]. Finally, second order (phonon) softening was computed as the ratio of the maximum predicted phonon frequency and the maximum DFT target phonon frequency.

We computed energy and force softening of all OAM models using the DFT energy and force values in the WBM high energy dataset[23] which are computed with DFT settings that match Materials Project and Matbench-discovery settings. Additionally, we ran single-point DFT calculations using the OMat24 settings described in Section 4.2 in order to compute energy and force labels to evaluate softening of OMat24 trained models.

Energy and force softening calculations are carried out each architecture (eSEN, equiformerV2, GRACE, SevenNet and MACE) by running model predictions using each of their ASE compatible calculator classes. Phonon softening is similarly calculated by running model inference via ASE calculators following the following protocol[49]:

1. Run a structure relaxation of the primitive cell with a convergence force threshold of 5 meV/Åusing the FIRE optimizer with FrechetCellFilter implemented in ASE[73].

2. Calculate force constants using the finite displacement supercell method implemented in Phonopy[74] using a displacement of 0.01 Å. The supercell for all structures were taken directly from the PBE MDR dataset.

3. Calculate the maximumum phonon frequency from phonon frequencies calculated at q-points commensurate with the supercell.

Data Availability

The OMat24 and subsampled Alexandria (sAlex) datasets, as well as the reference compounds and fitted MP2020-style Compatiblity corrections are available for download via Hugging Face [75]. VASP input generation files necessary to reproduce DFT calculations can be found in the FAIRCHEM-DATA-OMAT repository [76]. Source data for Figures 1-4 is available with this manuscript.

Code Availability

All model training and architecture code is available on the FAIRChem Github [77]. equiformerV2 and eSEN pretrained model checkpoints can be downloaded from Hugging Face [78]. Pretrained models are compatible with FAIRCHEM-CORE 1.10 [77].

Acknowledgements

We acknowledge helpful discussions with Miguel Marques (Ruhr University Bochum), Janosh Riebesell, Shyue Ping Ong (UCSD), Yi-Lun Liao (MIT), Tess Smidt (MIT), and Bowen Deng (U.C. Berkeley). The authors acknowledge valuable assistance from Kyle Michel and Lowik Chanussot (FAIR at Meta) in enabling and scaling the simulation datasets on Meta compute resources. We generated several of our figures using CRYSTAL-TOOLKIT [79] and PYMATVIZ [80].

Author Contributions

L.B.L., M.S., M.U., C.L.Z, and Z.U conceived, designed and planned the project. L.B.L. and M.S. generated input structures, ran DFT and processed all DFT calculations. L.B.L., M.S., X.F. and B.M.W. optimized hyperparameters and trained models. M.D. and M.G. implemented training and model inference infrastructure. A.R., M.U., and Z.U. supervised the work. L.B.L., M.S., X.F., B.M.W., C.L.Z, and Z.U wrote the paper. All authors edited the paper.

Competing Interests

Authors declare no competing interests.

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